Attachment 3 — Current Status Snapshot

> NLnet NGI Zero Commons Fund — Symphact application, 13th open call (deadline 2026-06-01).
> Snapshot date: **2026-05-23**. Repo: https://github.com/FenySoft/Symphact. Branch: `main`.

1. Milestone status at submission

Milestone	Status	Tests	Delivered
M0.1 Actor core primitives	✅ Shipped	30	`IMailbox` / `TMailbox`, `TActorRef`, `TActor<TState>`, `TActorSystem`
M0.2 ActorContext + inter-actor messaging	✅ Shipped	+16	`IActorContext` / `TActorContext`, `DrainAsync` with `MaxRounds`
M0.3 Supervision (let-it-crash)	✅ Shipped	+30	`ISupervisorStrategy`, `TOneForOneStrategy`, lifecycle hooks, hierarchy
M0.4 Scheduler + per-actor parallelism	✅ Shipped	+86	`IScheduler`, `TInlineScheduler`, `TDedicatedThreadScheduler`, `IMailboxSignal` / `TDotNetMailboxSignal`, `QuiesceAsync`
M0.5 Persistence — BCL-only slices	✅ Shipped	+44	`IJournal` + `TInMemoryJournal`, `ISnapshotStore` + `TInMemorySnapshotStore`
M0.5 Persistence — content-addressed	🚧 In progress	—	`TCasJournal` + `TCasSnapshotStore` (SHA-256 content-addressed) — *grant deliverable M1*
M0.6 Remoting + capability registry	⏳ Planned	—	`ITransport` + `TTcpTransport`, `TCapabilityRegistry` — *grant deliverable M2*
M0.7 CFPU integration demo	⏳ Planned	—	End-to-end demo over `FenySoft.CilCpu.Sim` — *grant deliverable M4*

M0.3 supervision, M0.4 scheduler and the M0.5 BCL-only slices shipped between v0.1 (2026-04-16) and this submission, ~65 hours of self-funded TDD work. These are explicitly out of grant scope — the grant funds the content-addressed production-grade journal, supervision lifecycle integration, remoting, capability registry, kernel + device actors, CFPU integration, formal-verification groundwork, and outreach.

2. Test suite — 186 green xUnit tests

```
$ dotnet test --nologo
Passed!  - Failed:     0, Passed:   120, Skipped:     0, Total:   120 - Symphact.Core.Tests.dll
Passed!  - Failed:     0, Passed:    22, Skipped:     0, Total:    22 -
Symphact.Platform.DotNet.Tests.dll
Passed!  - Failed:     0, Passed:    44, Skipped:     0, Total:    44 - Symphact.Persistence.Tests.dll
```

Test project	Count	Coverage
`Symphact.Core.Tests`	**120**	Actor core, scheduler, supervision, concurrency stress, mailbox signal
`Symphact.Platform.DotNet.Tests`	**22**	`TMailbox` (ConcurrentQueue MPMC), `TDotNetMailboxSignal` (AutoResetEvent WFI)
`Symphact.Persistence.Tests`	**44**	`TInMemoryJournal`, `TInMemorySnapshotStore` (append-only, replay, snapshot semantics)
Total	**186**	

CI runs the full suite on **Ubuntu / Windows / macOS** against .NET 10 SDK on every push (`.github/workflows/ci.yml`).

3. Code metrics

Layer	Files	Lines (.cs)
`src/Symphact.Core/`	18	runtime primitives, interfaces, supervision, scheduler
`src/Symphact.Persistence/`	6	journal + snapshot store interfaces + in-memory impls
`src/Symphact.Platform.DotNet/`	3	platform-specific .NET mailbox + signal impl
`src/Symphact.Platform.Cfpu/`	(stub)	awaiting CFPU F4 multi-core
`src/` total	**~27 files**	**~2 975 LOC**
`tests/` total	**~20 files**	**~3 890 LOC**
Test-to-runtime ratio		**~1.31×** (strict TDD discipline)

Build configuration (`Directory.Build.props`): `.NET 10`, `TreatWarningsAsErrors=true`, `GenerateDocumentationFile=true` — every warning is build-breaking, every public member requires XML docs.

4. Repository activity

- **Project start:** 2026-04-16 (initial repo scaffolding)
- **Snapshot date:** 2026-05-23
- **Calendar weeks:** ~5 weeks
- **Total commits:** 38
- **Focused TDD hours:** ~65 hours
- **Lines per hour:** ~106 (runtime + tests combined)

Recent commits (most recent first):

```
ac14a17 docs: NLnet pályázat v1.3 — post-Dennard paradigma-átírás + ördögügyvédi tényellenőrzés
396cbd2 feat: M0.5 második szelet — ISnapshotStore + TInMemorySnapshotStore (BCL-only)
1750e39 feat: M0.5 első szelet — IJournal + TInMemoryJournal (BCL-only)
dc236c4 docs: „1 core = 1 actor" helytelen állítás javítása + roadmap szétválasztás
db29cd1 docs: NLnet draft — PQC pontosítás LMS (NIST SP 800-208) hash-based signature-re
fc44d9b docs: NLnet pályázati draft v1.1 — beadás-kész 13. nyílt körre
e719fbe feat: M0.4 Scheduler + per-aktor parallelizmus
8e61fc1 docs: Armstrong–Hewitt hivatkozások hitelességi javítása a vision dokumentumokban
91eb5d0 docs: HMAC → CST modell átvezetés a Symphact dokumentációban
5f5443c feat: M0.3 Supervision + HAL platform architektúra
907bd46 feat: TActorRef(int SlotIndex) + CST capability modell — kaszkád átvezetés
```

5. Source-file inventory (runtime)

`src/Symphact.Core/` (18 files):

```
ECoreStatus.cs                IPlatform.cs              TActorRef.cs
ESupervisorDirective.cs       IScheduler.cs             TActorSystem.cs
IActorContext.cs              ISchedulerHost.cs         TDedicatedThreadScheduler.cs
IMailbox.cs                   ISupervisorStrategy.cs    TInlineScheduler.cs
IMailboxSignal.cs             TActor.cs                 TOneForOneStrategy.cs
IMailboxSignalProvider.cs     TActorContext.cs          TTerminated.cs
```

`src/Symphact.Persistence/` (6 files):

```
IJournal.cs        ISnapshotStore.cs        TInMemoryJournal.cs
TJournalEntry.cs   TSnapshotEntry.cs        TInMemorySnapshotStore.cs
```

`src/Symphact.Platform.DotNet/` (3 files):

```
TDotNetMailboxSignal.cs   TDotNetPlatform.cs   TMailbox.cs
```

6. Build verification

```
$ dotnet build Symphact.sln -c Debug
  Symphact.Core            -> bin/Debug/net10.0/Symphact.Core.dll
  Symphact.Persistence     -> bin/Debug/net10.0/Symphact.Persistence.dll
  Symphact.Platform.DotNet -> bin/Debug/net10.0/Symphact.Platform.DotNet.dll
  Symphact.Platform.Cfpu   -> bin/Debug/net10.0/Symphact.Platform.Cfpu.dll
  CounterActor             -> samples/CounterActor/bin/Debug/net10.0/Symphact.Samples.CounterActor.dll
  Symphact.Core.Tests            -> bin/Debug/net10.0/Symphact.Core.Tests.dll
  Symphact.Persistence.Tests     -> bin/Debug/net10.0/Symphact.Persistence.Tests.dll
  Symphact.Platform.DotNet.Tests -> bin/Debug/net10.0/Symphact.Platform.DotNet.Tests.dll
  Build succeeded.
    0 Warning(s)
    0 Error(s)
```

Zero warnings on a `TreatWarningsAsErrors=true` build with `GenerateDocumentationFile=true` — every public member is documented bilingually (`hu:` / `en:`) per repo convention.

6.1 Runnable end-to-end demo

```
$ dotnet run --project samples/CounterActor/CounterActor.csproj
Symphact CounterActor sample — sending 5 Increment + 2 Decrement + 1 Query

Query reply received: counter = 3
Final state via GetState<int>: 3
```

The `samples/CounterActor/` project (~80 lines, `Symphact.Samples.CounterActor`) demonstrates the full actor lifecycle: `TActorSystem(IPlatform)` creation, `Spawn<TActorType, TState>()`, `Send`, `IActorContext.Send` (capability-token-style reply), `Drain()`, and `GetState<TState>()`. See `samples/CounterActor/README.md` for the bilingual walkthrough.

7. Documentation state

Document	Version	Lines (en/hu)	Status
`docs/vision-en.md` / `vision-hu.md`	1.4	~750 / ~750	✅ Complete
`docs/roadmap-en.md` / `roadmap-hu.md`	1.0	~750 / ~750	✅ Complete
`docs/trust-model-en.md` / `trust-model-hu.md`	1.1	~250 / ~250	✅ Complete
`docs/boot-sequence-en.md` / `boot-sequence-hu.md`	—	—	✅ Complete
`docs/ddr5-memory-model-en.md` / `ddr5-memory-model-hu.md`	—	—	✅ Complete
`docs/actor-ref-scaling-en.md` / `actor-ref-scaling-hu.md`	—	—	✅ Complete
`docs/osreq-to-cfpu/osreq-001..006`	—	bilingual	✅ Active (007 superseded by CST model)
`docs/nlnet-application-draft-en.md` / `-hu.md`	1.3	302 / 350	✅ Submission-ready

Every document is bilingual (English + Hungarian) per repo convention. Cross-reference integrity is maintained across the two language sets.